Overnight Experiment Results

Adaptive Hybrid Semantic Research Assistant · 23AID472 Text Analytics · Batch A-17

Run 12–13 October 2026 on an RTX 4060 Laptop GPU (8 GB). Retrieval measured on the same 2,000 SQuAD v1.1 development questions used in the paper; end-to-end sample sizes are given per row. Every model is open-source and runs locally.

1. What the previous code did

The system as it stood before these experiments, taken from the committed code rather than from the paper's description.

Stage	What the previous code did	Where
Corpus	SQuAD v1.1 train + dev contexts de-duplicated to 20,958 passages, split into 23,288 chunks of at most 180 words with 40 words of overlap	src/data.py, src/chunking.py
Sparse index	BM25Okapi over the chunks, k1 = 1.5, b = 0.75, lower-cased alphanumeric tokens	src/retrieval_sparse.py
Dense index	all-MiniLM-L6-v2 embeddings, 384 dimensions, L2-normalised, FAISS IndexFlatIP (exact search)	src/retrieval_dense.py
Query analyser	Keyword rules label the question simple / comparison / multistep and set K = 5 / 10 / 15; string rules write up to two paraphrases	src/multiquery.py
First-stage search	Each of the 2–3 variants searched in both indexes, 50 candidates per list	notebook, Section 9
Fusion	Reciprocal Rank Fusion over the 4–6 lists, k = 60, keeping the top 30	src/fusion.py
Re-ranking	ms-marco-MiniLM-L6-v2 re-scores those 30 against the original question	src/rerank.py
Reader	RoBERTa-base reads the rank-1 chunk only; start and end taken as independent arg-maxes over the first 384-token window	src/qa_infer.py, src/pipeline.py

Three things in that list are worth separating out, because they are what the experiments changed:

• **K was computed and then ignored.** The retrieval study requested the top 20 and the end-to-end evaluation the top 3, and the reader used only the first result, so the complexity-based K never affected any published number.

• **The reader saw one passage.** Everything below rank 1 was retrieved, re-ranked and discarded unread.

• **The served API is not the evaluated pipeline.** app/api.py has no cross-encoder stage and searches whole passages rather than chunks, so the deployed demo is weaker than the system the paper measures.

2. What was changed, component by component

Stage	Baseline (current paper)	What we tried	Outcome
Dense embedder	all-MiniLM-L6-v2 (91 MB)	E5-base-v2, BGE-base, BGE-small	E5-base best; BGE worse than baseline
Cross-encoder	ms-marco-MiniLM-L6-v2 (91 MB)	ms-marco-MiniLM-L12-v2 (133 MB)	L12 wins: +1.27 MRR
Candidate depth	50 per list, re-rank 30	100 per list, re-rank 50	+0.65 Recall@10
Span decoder	arg-max, first window only	joint start/end, all windows	+0.65 EM / +0.78 F1
Reader	roberta-base (499 MB)	retrained, identical recipe	reproduced: 85.37 EM / 91.83 F1
Multi-query	rule-based string variants	flan-t5-small, trained paraphraser	no gain — see section 4

3. Retrieval: baseline versus best (2,000 questions)

Configuration	R@10	R@20	MRR	NDCG@10	ms/q
Baseline: MiniLM-L6 + L6, depth 50/30	96.25	96.65	85.25	87.97	164
+ depth 100/50	96.90	97.90	85.44	88.23	195
+ L12 re-ranker, depth 100/50	97.25	97.85	86.52	89.17	284
Best: E5-base + L12, depth 100/50	**97.50**	**98.35**	**86.79**	**89.42**	**296**
Improvement over baseline	**+1.25**	**+1.70**	**+1.54**	**+1.45**	**+132 ms**

Candidate depth buys recall (more correct passages reach the shortlist); the larger re-ranker buys ranking quality (MRR and NDCG), because re-ranking reorders a fixed candidate set rather than finding new passages.

4. The transformer for multi-query expansion

Variant source	Model	Usable variants	R@10	MRR	Gen cost
None (control)	—	—	96.25	85.25	0 ms
Rules (current system)	—	100%	96.25	85.19	0 ms
Transformer 1	google/flan-t5-small	496 / 4000 = 12%	96.05	85.14	23 ms
Transformer 2	chatgpt_paraphraser_on_T5_base	3569 / 4000 = 89%	95.45	84.78	19 ms

Both transformers lose to the rules. flan-t5-small answers the question instead of rewriting it, inventing facts that pollute the search ("What is the speed of light?" became "speed of light is 0.005 m/s"); only 12% of its output survived a filter requiring a question with no new numbers. The purpose-trained paraphraser produced excellent rewrites (89% usable) and still lost 0.80 Recall@10.

The reason is specific to this benchmark: SQuAD questions were written by crowdworkers looking at the passage, so the original wording already overlaps the target text more than any paraphrase. Every rewrite moves away from that optimum and dilutes the fused ranking.

5. End-to-end answering (RoBERTa-base reader)

Setting	Questions	EM	F1
Published result	100–200	71.00	72.34
Baseline retrieval, rank-1 only	200	66.50	69.77
Baseline retrieval + top-5 reading	200	72.00	76.24
Best retrieval, rank-1 only	500	75.60	78.41
Best retrieval + top-5 reading	**500**	**77.20**	**80.80**
Improvement over published		**+6.20**	**+8.46**

Better retrieval alone lifts rank-1 reading from 66.50 to 75.60 EM. Reading the five best passages and keeping the most confident span adds a further 1.60 EM. With the weaker baseline retriever, top-K was worth far more (+5.50 EM), because rank 1 was wrong more often.

6. How many passages the reader should read

K (passages read)	1	3	5	10
Exact Match	66.50	71.50	72.00	69.50
F1	69.77	74.85	76.24	74.11

Measured on 200 questions with identical retrieval. The gain peaks at K = 5 and reverses by K = 10: with more passages the reader eventually meets a confidently wrong answer that outranks the correct one. The paper's complexity classifier currently assigns 5 / 10 / 15, so the two larger values sit past the peak and should be reduced to the 3–5 range.

7. Reader checkpoint restored

	Published	Retrained tonight	Difference
Exact Match	85.38	85.37	−0.01
F1	91.74	91.83	+0.09
Training time	2.45 h (RTX 5060)	1.19 h (RTX 4060)	—

The fine-tuned checkpoint lost with the previous machine now exists again (models/roberta-base-squad1, 1.9 GB), which restores the FastAPI backend and the live demo. The recipe matched the original exactly: 8,301 optimizer steps, effective batch 32, 6% warm-up.
The published retrieval baseline was also reproduced identically in three independent runs (96.25 / 96.65 / 85.25 / 87.97), which confirms the numbers in the paper.